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Does expecting an external store impair memory for expository texts? Investigating cognitive offloading of text passages

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ABSTRACT

People often use external tools to support their memory, which can cause memory impairments for information expected to be externally accessible, effects often attributed to cognitive offloading. Memory offloading research has predominantly used facts or word lists, so little is known about the effects of store expectancy on longer texts. Three experiments investigated how anticipating external access to a passage influences memory for that passage. Participants read expository texts, one of which they expected to access during a later multiple-choice memory test, and one they did not expect to be able to access. In Experiment 1, participants had full control of their reading time. In Experiment 2, participants were required to spend a minimum time reading. In both experiments, participants spent less time reading the passage they expected to access at test. In Experiment 3, reading time was fully controlled. Participants also reported their subjective reading effort and gave metacognitive predictions of their test performance. Across all experiments, participants scored lower on the memory test when they anticipated access to the passage than when they did not, and reading time mediated the relationship between store expectancy and condition. These results indicate that offloading-related memory impairments extend to informational text passages, even when time spent reading the passages is controlled.

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The widespread availability of external storage systems has altered how and when people need to use memory to complete everyday tasks (for reviews, see Eliseev & Marsh, 2021; Marsh & Rajaram, 2019; Storm & Soares, 2024). Indeed, people report frequently using both digital and non-digital strategies to support or even replace their memory (Finley et al., 2018; Radnan et al., 2023). External stores of information, often digital devices like smartphones and computers, may circumvent the need to store and retrieve organically (Sparrow et al., 2011). In recent years, people report memorising few phone numbers; only a fraction of those stored externally in their phone contacts (Finley et al., 2018), presumably because people consider their phones responsible for "remembering" phone numbers for them. Ultimately, failing to learn phone numbers has few consequences for everyday living, unless a phone is lost. However, a much wider range of information is often available externally. Failure to remember more complex information, like the details of a story, could have more significant consequences in domains ranging from academic learning to social relationships.

Cognitive offloading refers to physical actions taken to decrease the demands or improve performance of a mental task (for review, see Risko & Gilbert, 2016). Memory is thought to be offloaded when a person believes that they will be able to access an external store to complete memory tasks. This offloading typically results in improvements to aided task performance, and impairments to unaided memory (for reviews, see Gilbert et al., 2023; Richmond & Taylor, 2025). In an illustrative example of cognitive offloading, Sparrow et al. (2011) investigated memory for trivia facts that participants believed to be saved on a computer or deleted. Participants then completed an unaided memory test for all facts. Participants' memory for "saved" facts was significantly impaired relative to their memory for "deleted" facts. The memory impairment for offloaded information has been replicated in a variety of contexts and with different materials (Richmond & Taylor, 2025). In tests of long-term memory, studies have shown memory impairments for offloaded information using sentence-based trivia facts (Schooler & Storm, 2021; Sparrow et al., 2011), word lists (e.g., Kelly et al., 2025; Kelly & Risko, 2022), and

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item-location pairs (Grinschgl, Papenmeier, et al., 2021). Other studies have also shown impairments to memory for information saved by taking a photograph relative to non-photographed information (Lurie et al., 2025; Lurie & Westerman, 2021; Soares & Storm, 2022; Wang et al., 2025), though there is dispute over whether offloading drives this impairment (Soares & Storm, 2018).

However, the effect of store expectancy on memory for longer, more complex materials like text passages has yet to be established. Some studies in which participants learn using internet search compared to being provided equivalent information have used longer texts (Fisher et al., 2022; Eliseev & Marsh, 2023). In these studies, participants likely believed they could later access information found through internet search to a greater extent than information they were provided by the experimenters. In one study, learning of expository texts accessed through internet search was impaired relative to learning by reading the same text without search, an effect the authors attributed to cognitive offloading (Fisher et al., 2022). In another study, participants explained common phenomena less effectively after searching online for the answers compared to participants who were provided the same text (Eliseev & Marsh, 2021). In both studies, store expectancy was confounded with other aspects of engaging in internet search; however, so it is difficult to interpret the extent to which such findings could be attributed to offloading and not to other aspects of engaging in internet search. Indeed, Eliseev and Marsh (2021) favoured the possibility that webpage previews shown by search engines drove these search-related impairment effects to a greater extent than offloading.

The *study effort hypothesis* suggests that learners expend less effort when they can rely on an external memory store than when they cannot, and that this change in effort drives the memory impairment associated with offloading (Kelly & Risko, 2019a, 2019b). Consistent with this account, Kelly and Risko (2022) had participants learn lists of words, typing each word to save to a list that appeared on the side of the computer screen. During these trust trials, participants took memory tests with the help of their self-generated lists. Prior to the final trial, half of the participants were warned that they would not have access to such a list, while half were not warned. Participants who expected access to the list at test spent less time studying, used fewer explicit study strategies, and performed worse on the final test compared to participants who expected to lose access. Moreover, these differences in study time and strategies between groups partially mediated the difference between store expectancy conditions representing the offloading-related memory impairment. Further supporting the view that these impairments occur during encoding, the store expectancy impairment persists when tested using recognition memory tests (Kelly et al., 2025). One might also expect that if participants who believe information is saved externally strategically put forth less cognitive effort, they would be metacognitively aware of this impairment, integrating the relative lack of effort into their predictions of future performance. Indeed, some work has shown that participants predict lower metacognitive expectations of unaided memory when they expect access to a store than when they do not expect access (Park et al., 2022).

Based on this study effort mechanism, there are good reasons to expect that offloading-related memory impairments extend to complex text information. Generally, fast reading can impair understanding of a text (Rayner et al., 2016). Previous work has shown that participants differentially allocate reading time based on the perceived value of information contained in text passages, resulting in longer reading times and better memory for high-value information (Cirilo & Foss, 1980). Subsequent work replicated this finding and also found that the effect of importance persisted even when reading time was controlled, suggesting that participants can also differentially engage study strategies or allocate cognitive effort when reading, even when they cannot control their time spent reading (Britton et al., 1986). So, if participants spend less time or put less effort into studying texts they expect later access to, as in previous studies using word lists (Kelly et al., 2025; Kelly & Risko, 2022), it seems likely that expecting store access would impair memory for those texts.

Reducing study effort also seems likely to influence important factors like interest, motivation, and reading goals. Interest in a text consistently predicts reading comprehension (for review, see Hidi, 2001), even when controlling for prior knowledge (Baldwin et al., 1985). Attention is thought to be an important component of interest (Ainley et al., 2002). For example, one study showed that interest in a text predicted motivation to read the text, which in turn negatively predicted lapses in attention while reading (Unsworth & McMillan, 2013). Store expectancy could also influence reading goals. More demanding goals often yield longer reading times (Yeari et al., 2015), behaviours consistent with deeper processing (Ishiwa et al., 2013; Van den Broek et al., 2001) and higher test performance (Coleman et al., 1997; Geiger & Millis, 2004; Guerrero & Wiley, 2021; Nestojko et al., 2014; Van den Broek et al., 2001). Reading with the expectation of an open resource test could constitute a particularly low-demand reading goal, resulting in poor learning. Indeed, Agarwal et al. (2011) found that expecting an open-book test reduced student participants' self-paced study time, indicating that anticipating external support can cause learners to strategically reduce their study effort.

However, there are reasons to expect that learning from text passages might engage different cognitive processes than learning word lists or sentences, which could protect against offloading-related memory impairments. First, text passages typically contain more interconnected information than lists of words or sentences. Kintsch (1982; 1998) argued that reading and comprehending

these passages requires a person to generate a hierarchical knowledge structure, in which higher-level comprehension is formed from learning of lower-level linguistic information. As a person reads, activation of this low-level information may occur as the higher-level structure forms, creating an integrated knowledge structure of strongly associated information (Graesser et al., 1994; Myers & O'Brien, 1998; Wolfe, 2005). This type of highly interconnected structure could make memory for texts more robust than for shorter or less complex stimuli. Associating words studied in a list by generating a story linking them together, adding narrative structure, can substantially enhance memory for those words (Bower & Clark, 1969). Activation of existing knowledge structures, or schemas, while reading can also enhance memory for a text (Bransford & Johnson, 1972). This kind of highly integrated schematic structure could mitigate offloading-related memory impairments. In fact, some have failed to find evidence of directed forgetting effects, which are often considered similar to offloading-related memory impairments (Eskritt & Ma, 2014; Kelly & Risko, 2019a, 2019b; Lu et al., 2020; Sparrow et al., 2011), using prose passages (Pastötter & Haciahmet, 2022). If similar mechanisms drive store-expectancy impairments and directed forgetting, expecting access to a passage might not cause the same kinds of memory impairments as expecting access to a list of words or sentences.

Second, reading a text may require deeper processing than attempting to learn a list of words or facts. Some have argued that store expectancy can determine the depth of information processing during encoding, with shallower encoding engaged for "saved" information (Burnett & Richmond, 2026). Reading and understanding a text requires a reader to process information based on semantic meaning, but word lists could be learned based on shallower feature-based processing (Craik & Lockhart, 1972; Craik & Tulving, 1975). Thus, processing "saved" text may not lead to the same shallow processing of information as processing "saved" word lists.

Third, even if store expectancy reduces reading time, reading time does not always predict learning from text. In correlational studies, reading time is influenced by a participant's reading ability, the complexity of the text, and participants' approach to reading the text itself (for review, see Wallot et al., 2014). However, some experimental manipulations that influence reading time do not always result in reliable differences in memory for a studied text (Yeari et al., 2015). It is possible that even if store expectancy reduces study time, it might not yield a corresponding memory impairment.

Finally, there are studies in the educational psychology domain demonstrating that having access to external stores does not always harm learning. Taking notes, which saves information, often benefits students' classroom learning (Kobayashi, 2006). Likewise, one study found that having access to printed slides during a lecture can even benefit learning (Marsh & Sink, 2010). It seems unlikely that students expected access to these notes or slide printouts during their tests, though. Another study found no reliable effects of the timing of lecture slide availability (before or after class) on learning, though all students in this study expected lecture materials to be externally available at some point (Babb & Ross, 2009). These results, though not directly analogous to studies showing a store expectancy impairment, suggest that access to an external store does not always impair learning and can sometimes even benefit learning.

The current study

The current study examined how expecting access to an expository text passage at encoding influenced later unaided memory performance. In Experiment 1, participants read passages and completed a memory test of the information contained in each passage. Participants were led to believe that they would have access to one passage at test but not another. In this first experiment, participants could also control how long they spent reading each passage. The study effort hypothesis claims that participants expend less cognitive effort when encoding information they believe to be saved externally than information not believed to be saved (Kelly & Risko, 2019a). Based on this hypothesis, we expected that participants would spend less time studying and have poorer memory for passages they expected access to at test relative to passages they expected no access to at test. All three experiments received ethical approval from Mississippi State University's Institutional Review Board through an exempt protocol (IRB-24-214). Participants were excluded from participating in more than one of the reported experiments.

Experiment 1

Method

Design

Experiment 1 was preregistered (https://aspredicted.org/73ki3g.pdf). Experiment 1 employed a 2-condition (Expected Access, Expected No Access) within-subjects design. Participants were asked to read expository text passages in anticipation of a memory test. During reading, participants in the Expected Access condition anticipated access to the passage during the memory test, while participants in the Expected No Access condition did not anticipate access to the passage at test. Passage assignment to condition and condition order were counterbalanced across participants.

Participants

A power analysis indicated we would need 90 participants to detect a pairwise difference with a small-medium effect size ($d = 0.30$) with 80% power and $\alpha = .05$; therefore, we aimed to collect a filtered sample of at least 90

participants. Participants were run online using Qualtrics in batches, so additional participants were collected to compensate for potential exclusions and non-submissions. A total of 124 undergraduates were recruited from Mississippi State University's Psychology Department participant pool in exchange for partial course credit. Nine participants were excluded: one because they reported that they did not follow directions at the end of the study by using outside sources at test, speeding through, or multitasking during the study, and eight participants for answering multiple choice questions on all memory tests at a rate below chance (25%). This left us with a final sample of 115 undergraduate participants ($M_{age} = 20.82$, $SD = 4.15$). In this sample, 60% were women, 38% were men, and 2% were non-binary, third gender, or agender.

Materials

Full materials are available through the Open Science Framework (https://osf.io/enhqt/). The materials consisted of four educational passages from National Geographic (Andrews, 2024; Arnold, 2023; Méndez, 2024; Sartore, 2024). These passages focused on various unrelated topics, including a Mars volcano, spring cleaning, penguin sleep, and the history of underwear, ($M_{word\ count} = 724.7$, $SD = 94.6$). We selected expository text passages with estimated reading times of about 5 min each. We selected topics that were somewhat esoteric to limit participant prior knowledge. An online Flesch Kincaid Calculator (Flesch Kincaid Calculator, n.d.) indicated that each text was at a high school or college level (Mars volcano: Grade Level: 12, Ease Score: 43.3; spring cleaning: Grade Level: 11, Ease Score: 51.4; penguin sleep: Grade Level: 10.2, Ease Score: 53.4; history of underwear: Grade Level: 10.3, Ease Score: 53.1). Given that the participants were university students, these texts were appropriate for them to read based on their level of education.

There were 10 multiple-choice memory questions for each passage, totalling 40 questions. The questions were constructed such that they asked participants to recall explicitly stated details from sentences throughout the passages. For example, the following is an excerpt from the Mars Volcano passage: "At almost 30,000 feet tall (a little higher than Everest), and perhaps 280 miles long at its base … ". A question for this passage was: "How long was the potentially discovered volcano at its base?" with the correct answer being *280 miles*. The following is an example from the penguin sleep passage: "When nesting, these Antarctic birds take four-second-long 'microsleeps', a strategy that allows parents to keep constant watch over vulnerable eggs and chicks, all while racking up 11 hours of total sleep a day, according to a new study". A question for this passage was: "Chinstrap penguins take 4-second microsleeps, which add up to … " with the correct answer being *11 hours*. Each question appeared with the correct answer option as well as 3 lures in a random order.

Procedure

The experiment was conducted entirely online and through Qualtrics. After providing consent, participants were informed that they would read a few short passages in anticipation of answering questions about the information covered by the reading. They were asked to acknowledge that they would not take any notes or use external sources to answer the questions. Participants were required to type a statement agreeing to those terms before beginning the first phase. Experiment 1 was entirely self-paced, including all reading and testing phases, and participants took a median of 24 min to complete the study.

The first phase was a trust phase designed to show participants that they could reliably expect access to passages as instructed during the study. Prior work indicates that participants must trust the reliability of a store for the offloading-related memory impairment to be observed (Schooler & Storm, 2021). In the trust phase, participants read two unrelated passages. Prior to reading the first passage about a volcano on Mars, participants were told that they would have access to the first passage, and that it would be available to reference when answering questions later. They were told that it was still important for them to read the entire passage, but that they would have access to it later if they needed it. For the next passage about spring cleaning, participants saw instructions that this passage would *not* be available for later reference, and that it was important for them to read this passage in its entirety as well.

After reading both passages, participants answered ten multiple choice questions about the Mars volcano. Each question appeared on a separate page alongside the full text of the article, consistent with participants' expectations. They then answered ten questions about the spring-cleaning passage, which appeared without the passage available. For half of participants, passage assignment to condition and condition order was flipped such that participants did not expect access to the first passage about the Mars volcano and expected access to the second about spring cleaning.[1]

Participants then moved on to the experimental phase, which was identical to the practice phase, except for one critical difference. While participants were still told they would have access to a passage before reading, which constituted the Expected Access condition, they did not have access to either passage during the final test in either condition. Self-paced reading time was measured as the time spent on the pages containing each passage. In this experimental phase, participants saw two new unrelated passages: the first about penguin sleep and the second about the history of underwear. Half of participants were told that they would have access to the first passage at test, and half of participants were told that they would have access to the second passage at test. The order conditions were run and passage assignment to condition

matched the order given during the trust phase for each participant. After reading both passages, participants were told that they would answer questions about each passage, but that contrary to previous instructions, they would not receive access to either passage. They then completed both tests in the order the passages appeared, with each question appearing on a separate page. Participants then answered demographic questions, were debriefed, and then dismissed from the experiment with credit.

Results and discussion

Memory performance

Memory test performance was scored using the multiple-choice key. As expected, participants performed near ceiling on the test regarding the Mars volcano, which they had the full text available for during the test (M = 93.8%, SD = 13.5%). Participants' performance was much lower for the spring cleaning passage (M = 54.4%, SD = 24.3%), in which they completed the test unaided, $t(114) = 17.14$, $p < .001$, $d = 1.60$, 95% CI_d [1.32, 1.87].

In the experimental phase, a paired-samples t-test indicated that memory performance in the Expected Access condition (M = 43.7%, SD = 24.9%) was significantly lower than in the Expected No Access condition (M = 52.2%, SD = 24.9%), $t(114) = 3.38$, $p < .001$, $d = 0.32$, 95% CI_d [0.13, 0.50] (see Figure 1).[2] This finding is consistent with previous research demonstrating offloading-related memory impairments. The data were also analysed using a Generalied Linear Mixed Effects model using the *lme4* (1.1-33) package in *R* (Bates et al., 2015). The model was constructed to address the influence of Store Condition on memory performance while controlling for random effects of participants, fixed effect of passage assignment, and items used to test memory. The model included fixed effects of Store Condition and Passage, and random intercepts and slopes for both participants and items (`Mod1 <- glmer(Memory Performance ~ Store Condition + Passage + (1 + Store Condition|PID) + (1 + Store Condition|Item), family = "binomial")`). There was a significant effect of Store Condition such that the Expected Access condition performed significantly worse relative to the Expected No Access condition ($\beta = -0.43$ [0.13], $z = -3.18$, $\chi^2 = 8.91$, $p = .001$). The effect of Passage was not significant ($\beta = -0.01$ [0.15], $z = -0.07$, $\chi^2 = 0.00$, $p = .946$).

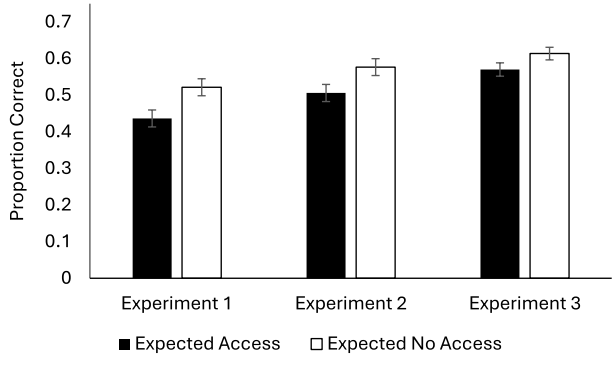


Figure 1. Memory performance as a function of source expectancy in experiments 1–3. *Note.* Error bars indicate the standard error of the mean.

Reading time

Participants also spent significantly less time reading the passages for which they Expected Access ($M = 105.96$ *s*, $SD = 214.51$) compared to passages for which they Expected No Access ($M = 184.80$ *s*, $SD = 226.80$), $t(114) = 4.81$, $p < .001$, $d = 0.45$, 95% CI_d [0.26, 0.64]. A separate Linear Mixed Effects model examined the influence of Store Condition on Reading Time, controlling for participant level variability. The model included fixed effects of Store Condition and Passage as well as a random intercept for participants (`Mod1RT <- lmer(Reading Time ~ Store Condition + Passage + (1|Participant))`). There was a significant effect of Store Condition such that participants in the Expected Access condition spent significantly less time reading than participants in the Expected No Access condition ($\beta = -0.35$ [0.02], $t(2183) = -20.90$, $\chi^2 = 398.63$, $p < .001$). The effect of Passage was not significant ($\beta = 0.00$ [0.02], $t(2183) = 0.18$, $\chi^2 = 0.03$, $p = .856$).

Mediation: reading time

A mediation analysis (Tingley et al., 2014) was conducted to examine reading time as a mediator between Access Expectancy (treatment) and memory performance (outcome). This analysis compares a mediator model (`glmer(Memory Performance ~ Store Condition + Reading Time + (1 + Store Condition |Participant), data = data, family = binomial)`), and an outcome model (`lmer(Reading Time ~ Store Condition + (1|Participant), data = data)`). Given that Store Condition significantly predicted Reading Time (a-path; $\beta = -0.35$ [0.02], 95% CI_β [−0.38, −0.32], $t(2184) = -20.91$, $\chi^2 = 398.60$, $p < .001$), and that Reading Time significantly predicted Memory Performance when controlling for Store Condition (b-path; $\beta = 0.49$ [0.08], 95% CI_β [0.33, 0.65], $z = 6.00$, $\chi^2 = 35.32$, $p < .001$), the two variables can be considered in a follow-up mediation analysis. There was a significant total effect of Store Condition on test performance, $\tau = -0.08$, 95% CI_τ [−.13, −.03], $p < .001$. There was also a significant average causal mediation effect of Store Condition on test performance, $\delta = -0.04$, 95% $CI\delta$ = [−.05, −.02], $p < .001$. The direct effect of Store Condition on test performance trended towards statistical significance, $\zeta = -0.05$, $CI\zeta$ = [−.09, −.00], $p = .076$. These results leave some ambiguity regarding full or partial mediation, but suggest that Store Condition significantly influenced reading time, which significantly influenced memory test performance.

Experiment 2

Experiment 1's findings suggest that store expectancy impairments extend to expository text passages. We also found that participants spent less time reading passages they expected to access later than passages they did not expect to access, and that reading time mediated the effect of store condition on memory performance. These findings are consistent with the study effort hypothesis and with previous work showing that participants spend less time studying words they believe they will have access to on a later test than words they believe will be inaccessible (Kelly et al., 2025; Kelly & Risko, 2022).

One concern about Experiment 1 was that reading times were quite variable across participants, and many participants' reading times indicated that they did not read the passages provided in their entirety. A recent meta-analysis suggests that typical English silent reading speed is 238 words per minute (Brysbaert, 2019). The experimental passages presented to participants averaged 772 words, suggesting an expected average reading time of about 194 *s*. Despite this, 51 participants spent less than 10 *s* on the page containing the passage that they expected to access later, with 22 participants spending less than 10 *s* on the page containing the passage they did not expect to access again. While consistent with the predictions of the study effort hypothesis, this finding supports the possibility that the effect observed in Experiment 1 was driven largely by participants who chose not to read the Expected Access passage. Experiment 2 addressed this possibility by requiring participants to spend a minimum amount of time on the page containing each passage. Experiment 2 employed a timer for each passage which disallowed participants from advancing from the page containing each passage until at least 90 s had elapsed. This timer was intended to ensure that participants could not skip past the passages entirely.

Method

Experiment 2 used the same design and procedure as Experiment 1, except where noted. Experiment 2 was preregistered (https://aspredicted.org/kz9a2m.pdf).

Participants

A total of 101 participants were recruited. Two participants were excluded for answering multiple choice questions on all memory tests at a rate below chance (25%). After exclusions, the sample consisted of 99 undergraduates (M_{age} = 19.5, SD = 4.1), 80% women, 16% men, and 3% non-binary, third gender, or agender. This exceeded our preregistered stopping plan slightly, again because additional participants were run in batches to account for exclusions.

Procedure

The procedure for Experiment 2 was identical to that of Experiment 1, except that a 90 s timer was added to the page containing each passage, which ensured that participants could not advance until at least 90 s passed. Participants could not see the timer on the screen; the button to advance only appeared after 90 s. A minimal time of 90 s was selected because it was less than half the reading time expected based on average English silent reading speeds (Brysbaert, 2019). As such, we hoped to stop participants from clicking past the passage entirely while making it unlikely that even fast readers would finish reading the passage and be unable to advance, which could encourage rereading.

Results and discussion

Memory Performance

As in Experiment 1, participants scored highly on the test for which the passage was available (M = 96.1%, SD = 9.2%), and much lower on the test they took unaided (M = 61.9%, SD = 22.5%) during the trust phase, $t(98)$ = 15.75, $p < .001$, d = 1.58, 95% CI_d [1.29, 1.88]. In Experiment 2, 13 participants scored below chance in the Told Store condition and 9 participants scored below chance in the Told No Store condition.

Replicating the results of Experiment 1, a paired-samples t-test indicated a significant effect of store expectancy on memory performance such that when participants Expected Access (M = 50.6%, SD = 23.3%) to the passage at test, memory performance was significantly impaired relative to when participants Expected No Access (M = 57.7%, SD = 22.9%) at test, $t(98) = 3.46$, $p < .001$, d = 0.35, 95% CI_d [0.14, 0.55] (see Figure 1).[3]

Using an approach identical to Experiment 1, the data were also analysed using a Generalized Linear Mixed Effects model to address the influence of Store Condition on memory performance. There was a significant effect of Store Condition such that the Expected Access condition performed significantly worse relative to the Expected No Access condition ($\beta = -0.37$ [0.11], $z = -3.45$, $\chi^2 = 11.04$, $p < 0.001$). The effect of Passage was not significant ($\beta = -0.06$ [0.37], $z = -0.17$, $\chi^2 = 0.03$, $p = .863$).

Reading time

Despite Experiment 2's implementation of a minimal reading time, when participants Expected Access ($M_{seconds}$ = 213.62, SD = 165.15) to the passages at test, they again spent significantly less time reading the passages relative to when participants Expected No Access ($M_{seconds}$ = 253.67, SD = 214.23) to the passages, $t(98) = 2.14$, $p = .035$, d = 0.22, CI_d [0.15, 0.41]. As in Experiment 1, a separate Linear Mixed Effects model examined the influence of Store Condition on Reading Time. There was, again, a significant effect of Store Condition such that participants

in the Expected Access condition spent significantly less time reading than participants in the Expected No Access condition ($\beta = -0.20$ [0.02], $t(1880) = -9.54$, $\chi^2 = 89.29$, $p < 0.001$). The effect of Passage was not significant ($\beta = 0.01$ [0.02], $t(1880) = 0.68$, $\chi^2 = 0.46$, $p = 0.49$).

Mediation: reading time

Using an identical procedure as Experiment 1, a mediation analysis was conducted to examine reading time as a mediator between Store Condition (treatment) and memory performance (outcome). This analysis compares a mediator model (`glmer(Memory Performance ~ Store Condition + Reading Time + (1 + Store Condition|Participant), data = data, family = binomial)`), and an outcome model (`lmer(Reading Time ~ Store Condition + (1|Participant), data = data)`). Given that Store Condition significantly predicted Reading Time (a-path; $\beta = -0.20$ [0.02], 95% CI_β [−0.16, −0.24], $t(1880) = -9.54$, $\chi^2 = 88.83$, $p < .001$), and that Reading Time significantly predicted Memory Performance when controlling for Store Condition (b-path; $\beta = 0.28$ [0.08], 95% CI_β [0.13, 0.43], $z = 3.67$, $\chi^2 = 12.67$, $p = .005$), the two variables can be considered in a follow-up mediation analysis. There was a significant total effect of Store Condition on test performance, $\tau = -0.07$, CI_τ [−.11, −.03], $p < .001$. There was also a significant average causal mediation effect of Store Condition on test performance, $\delta = -0.01$, $CI\delta = [-.02, -.00]$, $p = .002$ and a significant direct effect of Store Condition on test performance, $\zeta = -0.06$, $CI\zeta = [-.10, -.02]$, $p = .006$. These results suggest that reading time partially mediated the effect of Store Condition on memory performance.

Experiment 3

Experiment 2 replicated Experiment 1 as well as previous research demonstrating memory impairments when participants expect access to information at test. Experiment 2 also replicated the finding in Experiment 1 that participants spent less time reading when they expected access to the passage at test. Experiment 3 sought to investigate whether this store expectancy impairment persisted even when reading time was controlled entirely. As such, we could ensure that all participants were equally exposed to the passages. In Experiment 3, which took place in-person in the lab, participants listened to a text-to-audio recording of each passage while reading along. This design controlled for reading time and ensured that participants processed all parts of each text, at least to the extent that they heard the words. Participants could, therefore, only differentially allocate cognitive effort across the passages by differentially employing study strategies or allocating attention, not by varying the time they spent reading.

As such, we also had participants rate their subjective effort in Experiment 3. The study effort hypothesis suggests that store expectancy impairments occur because participants put forth less effort when they can rely on an external store to remember information for them than when they cannot. We therefore expected that participants would report exerting less effort when they expected later access to the passage than when they expected no access.

Finally, Experiment 3 also included metacognitive judgements of memory administered just before the test phase. We were interested in whether participants would be metacognitively aware of any memory effects associated with source expectancy. Previous work suggests that metacognitive judgements of memory ability can drive participants' decisions to offload memory (Hu et al., 2019; Risko & Dunn, 2015; cf. Grinschgl, Meyerhoff, et al., 2021). Indeed, some have argued that metacognitive beliefs about performance and effort may drive offloading decisions to a greater extent than actual performance and effort (Dunn & Risko, 2016). Though participants in the current study could not control their offloading behaviour, we were interested in their metacognitive expectations of their own performance after learning that they would be tested without being able to access the passage as expected. One might expect that if participants intentionally allocated little effort towards learning the passage they expected access to, they would be metacognitively aware that their memory was relatively impaired for that passage in its absence.

Method

Experiment 3 used the same design and procedure as Experiment 2, except where noted. Experiment 3 was pre-registered (https://aspredicted.org/ii42k7.pdf).

Design

Because reading time was controlled in Experiment 3, it was not measured. Experiment 3 also included metacognitive judgements and subjective ratings of effort. Participants gave metacognitive judgements of their anticipated performance and ratings of effort just before the final multiple-choice test. These subjective ratings of effort were then used in place of reading time as the mediator in a mediation analysis similar to Experiments 1 and 2.

Participants

A total of 157 undergraduates participated in Experiment 3. An additional manipulation check was included in Experiment 3 such that participants were asked at the end of the study to report the extent to which they believed that they would have access to the text for one of the passages on the final test. Prior work suggests that belief in the reliability of a saving process is important for observing store expectancy related memory

impairments (Schooler & Storm, 2021). Participants who reported that they did not "agree" or "strongly agree" that they expected access to one of the passages were excluded from the final analysis, amounting to 33 exclusions.[4] No participant reported the use of outside sources or other issues with their participation. The final sample consisted of 124 participants, 53% women, 44% men, 1% non-binary, third gender, or agender, and 2% who chose not to report or self-described ($M_{age} = 19.2$, $SD = 3.5$). As in previous studies, we aimed to collect data from at least 90 participants, after exclusions. Additional participants were scheduled to account for the additional exclusion criteria and no-shows. Because data collection took place near the end of an academic semester, we opted to collect data until the end of the semester and run all participants scheduled.

Procedure

The procedure for Experiment 3 was similar to that of Experiments 1 and 2. However, participation took place in person in a university laboratory. Participants completed the study in groups of up to 6 participants at a time. Each participant was seated at a separate partitioned computer desk. Once seated, participants began the study on Qualtrics as in previous studies. The study appeared as in Experiments 1 and 2, except that, for all passages, participants heard audio which read each passage through a provided set of headphones. The audio was generated using a text-to-speech tool which used a feminine-pitched voice and read at an average rate of 153 words per minute.

Otherwise, the experiment proceeded as in previous experiments until just before participants were presented with the tests during the experimental phase. As in previous experiments, before seeing any test questions, participants were informed that despite previous instructions, they would not have access to either passage at test. They were then asked to give metacognitive judgements using the following question:

> What percentage (out of 100) of the questions on a multiple-choice test of your memory for each topic do you expect to get correct? Keep in mind that you will not have access to either passage on the test and will answer from memory alone

They used sliding scales that appeared labelled with each topic (penguin sleep and history of underwear) in the order in which participants read them. To collect ratings of effort, participants were then asked the following question: "From the audio passages you just listened to, rate how closely you paid attention to each audio passage. 0 indicates no attention paid at all, 100 indicates the maximum amount of effort you could have put forward". For both questions, participants answered a qualitative short-answer question asking them to explain why they rated the passages differently for either scale if they had done so. Participants always provided metacognitive ratings before ratings of subjective effort. Participants then completed both tests as in the previous studies.

The final change was that, after completing the tests and prior to reporting demographics, participants completed an additional manipulation check. Participants were asked to rate their agreement with the statement "On the final test, I expected to have access to the manuscript for one of the passages", which they answered using a 1-5 Likert scale ranging from *strongly agree* to *strongly disagree*.

Results and discussion

Again, in the trust phase, participants scored very highly on the test during which they had access to the passage ($M = 98.4\%$, $SD = 4.5\%$), and much lower on the test they took unaided ($M = 62.2\%$, $SD = 18.5\%$), $t(119) = 20.98$, $p < .001$, $d = 1.92$, 95% CI_d [1.61, 2.22]. In Experiment 3, only 1 participant scored below chance in the Told Store condition.

Memory performance

Consistent with the findings of Experiments 1 and 2, a paired-samples t-test indicated a significant memory impairment when participants Expected Access ($M = 57.0\%$, $SD = 20.2\%$) compared to when participants Expected No Access to the passage at test ($M = 61.4\%$, $SD = 19.2\%$), $t(123) = 2.40$, $p = .018$, $d = 0.22$, 95% CI_d [0.04, 0.39] (see Figure 1).[5] The same Generalized Linear Mixed Effects model was also used as in Experiments 1 and 2, including a fixed effect of Store Condition and random intercepts and slopes for both participants and items. The model indicated a marginally significant effect of Store Condition such that the Expected Access condition tended to perform worse relative to the Expected No Access condition ($\beta = -0.24$ [0.13], $z = -1.95$, $\chi^2 = 3.44$, $p = .051$). Though statistically ambiguous, this result directionally aligned with the results of the t-test. The effect of Passage was not significant ($\beta = -0.12$ [0.45], $z = -0.27$, $\chi^2 = 0.07$, $p = 0.79$).

Metacognitive judgements

Metacognitive judgements were then assessed. In their raw judgements, participants were significantly more confident when they had previously Expected Access ($M = 74.2\%$ $SD = 24.5\%$) than when they Expected No Access ($M = 62.1\%$, $SD = 22.6\%$) to the passages, $t(123) = 8.03$, $p < .001$, $d = 0.72$, 95% CI_d [0.52, 0.92]. A separate Linear Mixed Effects model examined the influence of Store Condition on metacognitive judgements. The model included a fixed effect of Store Condition and Passage and a random intercept for participants `(Mod3MCJ <- lmer(MetacognitiveJudgements ~ Store Condition + Passage + (1|Participant)))`. The model also indicated a significant effect of Store Condition such that

participants in the Expected Access condition were significantly more confident than participants in the Expected No Access condition ($\beta = 12.20$ [0.35], $t(2279) = 34.55$, $\chi^2 = 960.76$, $p < .001$). The effect of Passage was also statistically significant ($\beta = 1.04$ [0.35], $t(2279) = 2.95$, $\chi^2 = 8.72$, $p = 0.003$), indicating that participants provided higher metacognitive judgements for the history of underwear passage relative to the penguin sleep passage.

To assess metacognitive accuracy, deviation scores were computed as the difference between participants' raw metacognitive judgement and their actual performance expressed as a percentage, such that positive values indicate overestimation and negative values indicate underestimation of actual performance. Participants were significantly more overconfident in their judgements when they Expected Access ($M = 17.19\%$, $SD = 2.78\%$) relative to when they Expected No Access ($M = 0.74\%$, $SD = 2.15\%$), $t(123) = 6.82$, $p < .001$, $d = 0.61$, 95% CI_d [0.42, 0.80].

Given these findings, we checked that participants understood that they would not have access to either passage on the final test by reviewing participants' explanations for why they provided different ratings for the readings, if they had done so. Despite participants having been reminded that they would not have access to either passage on the test in the question itself, 53 participants gave answers indicating that they expected to have access to one of the passages as instructed. As such, we re-ran the metacognitive judgement analyses excluding these participants (remaining $N = 71$) and surprisingly observed the same pattern of results. Participants reported significantly higher expected performance for passages they had initially expected to have available at test ($M = 64.2\%$, $SD = 24.4\%$) than passages they did not expect to have available ($M = 57.6\%$, $SD = 24.5\%$), $t(70) = 3.83$, $p < .001$, $d = 0.46$, 95% CI_d [0.21, 0.70]. With respect to metacognitive accuracy, participants were more overconfident in their performance when they Expected Access ($M = 5.4\%$, $SD = 29.2\%$) than when they Expected No Access ($M = -2.4\%$, $SD = 24.1\%$), $t(70) = 2.52$, $p = .014$, $d = 0.30$, 95% CI_d [0.06, 0.54].

Subjective effort ratings

Metacognitive judgements significantly predicted subjective effort ratings, as evidenced by positive correlations between metacognitive judgements and subjective effort in both the Expected Access, $r(122) = .50$, $p < .001$, 95% $CI_r = [.36,.62]$, and Expected No Access, $r(122) = .63$, $p < .001$, 95% $CI_r = [.51, .73]$, conditions.

A t-test indicated that subjective ratings of effort did not significantly differ when participants Expected Access ($M = 75.2\%$, $SD = 20.7\%$) compared to when they Expected No Access ($M = 76.7\%$, $SD = 21.6\%$), $t(123) = -1.19$, $p = .237$, $d = 0.11$, 95% CI_d [−0.07, 0.28]. Participants' qualitative responses about why they rated their effort differently between the two conditions, if they had done so, were reviewed. A total of 78 participants reported different ratings between the two conditions. Of these, only 23 participants (29.5%) gave responses mentioning that their efforts to pay attention differed between passages because they had believed one of the passages would be available at test.

A separate Linear Mixed Effects model examined the influence of Store Condition on subjective effort. The model included a fixed effect of Store Condition and Passage and a random intercept for participants (`Mod3SE <- lmer(SubjectiveE□ort ~ Store Condition + Passage + (1|Participant), data =` data)). Contrary to the results indicated by a t-test, the model indicated a significant effect of Store Condition such that participants in the Expected Access condition reported significantly less effort than participants in the Expected No Access condition ($\beta = -0.08$ [0.01], $t(2279) = -5.49$, $\chi^2 = 29.96$, $p < .001$). The effect of Passage was not significant ($\beta = 0.00$ [0.01], $t(2279) = 0.18$, $\chi^2 = 0.03$, $p = 0.86$).

Mediation: subjective effort

Because reading time was fully controlled for in Experiment 3, another mediation analysis was conducted to examine subjective effort as a mediator between Store Condition (treatment) and memory performance (outcome). This analysis compares a mediator model (`glmer(Memory Performance □ Store Condition + Subjective E□ort + (1 + Store Condition | Participant), data =` data, `family =` binomial)), and an outcome model (`lmer(Subjective Effort ~ Store Condition + (1|Participant), data =` data)). Given that Store Condition significantly predicted Subjective Effort (a-path; $\beta = -0.07$ [0.01], 95% CI_β [−0.05, −0.11], $t(2279) = -5.49$, $\chi^2 = 29.93$, $p < .001$), and that Subjective Effort significantly predicted Memory Performance when controlling for Store Condition (b-path; $\beta = 0.20$ [0.06], 95% CI_β [0.09, 0.32], $z = 3.55$, $\chi^2 = 11.97$, $p < .001$), the two variables can be considered in a follow-up mediation analysis. There was a significant total effect of Store Condition on test performance, $\tau = -0.04$, CI_τ [−.08, −.00], $p = .024$. There was a trend towards a causal mediation effect of Store Condition on test performance, $\delta = -0.00$, $CI\delta = [-.01, -.00]$, $p = .090$, but it failed to reach statistical significance. There was a significant direct effect of Store Condition on test performance, $\zeta = -0.04$, $CI\zeta = [-.08, -.00]$, $p = .032$. These results indicate a trend consistent with subjective effort partially mediating the effect of Store Condition on memory performance, but this trend was statistically ambiguous and should be interpreted with caution.

General discussion

Across three experiments, we observed that participants' memory was impaired for an expository text passage when they expected access to the passage at test

compared to when they did not expect access. This pattern aligned with previous research finding offloading-related memory impairments with word lists, trivia facts, and item-location pairs (Grinschgl, Papenmeier, et al., 2021; Kelly et al., 2025; Kelly & Risko, 2022; Schooler & Storm, 2021; Sparrow et al., 2011). Even though the effect was nominally smaller across experiments, the effect persisted in Experiment 3 even when participants were essentially required to process the text. These findings indicate that text passages are not exempt from offloading-related memory impairments even though text passages include interconnected, meaningful schematic information that requires semantic processing to comprehend.

In Experiments 1 and 2, participants also spent less time reading passages they expected to have access to at test, and reading time at least partially mediated the effect of store expectancy on memory performance. These findings are consistent with both the predictions of the study effort hypothesis and findings examining memory performance and study time using word lists (Kelly et al., 2025; Kelly & Risko, 2022). However, when reading time was controlled in Experiment 3, participants' reports of subjective effort did not predict their memory performance as clearly. We could detect reliably higher self-reported effort when participants expected no access than when they expected access, only when controlling for participant-level variance. Subjective effort also did not reliably mediate the effect of condition on memory performance, as reading time did. It is possible that factors other than conscious study effort contributed to the memory impairment observed in Experiment 3. That said, the ratings provided were subjective, so participants may not have accurately reported the cognitive effort they put towards learning the passages. Compounding this issue, participants listened to the passages in Experiment 3, which may have given participants the impression that they were attending well even when their attention strayed. Future work could circumvent these issues by collecting subjective ratings of effort during self-paced reading or using other approaches of probing attention, like eye tracking, which can detect evidence of mindless reading (Reichle et al., 2010).

Interestingly, in Experiment 3, participants also reported higher confidence in their memory for the passages they initially expected access to at test compared to the passages they did not. As a result, participants were more overconfident in their metacognitive calibrations when they expected access. This finding was observed even when participants who gave qualitative responses, indicating that they still expected access to one of the passages at test, were excluded from the analysis. This finding of relative overconfidence in memory for information learned, with the expectation of later access, aligns with findings that participants who use internet search can develop overconfidence in their own memory and other cognitive abilities (Eliseev & Marsh, 2021; Fisher et al., 2015, 2022; Ward, 2021; cf. Bittner & Storm, 2026). One possibility is that expecting later access to a passage influenced participants' subjective experience of learning the passage, such that listening to the passage felt easier or more fluent. Such feelings of processing fluency have been consistently shown to inflate metacognitive expectations of memory (for review, see Schwartz & Jemstedt, 2021). That said, given that many participants' qualitative responses indicated that they had not noticed the repeated warnings that none of the passages would be available at test, there remains the possibility that some proportion of participants who did not notice these warnings failed to indicate misunderstanding in their qualitative responses. Responses from such participants could have driven the observed results. As such, this finding should be considered preliminary, and future work should further investigate the metacognitive consequences of store expectancy.

The store expectancy effects on memory observed in the current study were robust, but small. These findings contrast with other work showing that store expectancy impairments on memory can be quite large. For example, Kelly and Risko (2022) observed large impairments using word lists (E1a: $d = 1.10$, E1b: $d = 0.71$), and Schooler and Storm (2021) observed large impairments using sentence-based trivia facts (E1: $d = 0.75$, E2: $d = 0.89$). Differences in the participant populations, formats, or specific design choices used across experiments could have contributed to these cross-experiment differences. Kelly and Risko (2022) employed three rounds of trust trials in which participants were shown that they would have access to saved words on a memory test administered at the end of each trial, whereas these experiments included one trust phase. These trust trials appear to be critical for observing offloading-related memory impairments (Schooler & Storm, 2021). Though we limited our analysis in Experiment 3 to participants who agreed that they had believed they would have access to the passage they expected, it seems likely that this belief would have been stronger if more trust trials had been employed. It is possible, though, that memory for text passages is partially protected from store expectancy impairments relative to memory for shorter, less complex or integrated stimuli. Future work could examine this possibility by directly comparing effects of store expectancy on memory for a larger variety of stimuli within the same experiment.

The current study used expository text materials like those often used in educational settings. Our results align with other work suggesting that anticipating access to external resources during a test can impair learning (Agarwal et al., 2011). As such, our results favour closed-book tests over open-resource tests in which students expect access to a saved resource. That said, the current study focused specifically on memory-based representations; the questions required recalling surface-based text representations and not representations that required inference or understanding. It is an empirical question

whether these findings would extend to inference-based or other comprehension questions commonly employed in learning environments. It is also unclear how the source expectancy effects observed in the current study would be affected by repeated engagement with texts, restudy, or tests of many or even longer texts as are typically assigned in university classrooms. A recent large meta-analysis showed that willingness to engage cognitive effort predicts academic achievement (Liu & Nesbit, 2024). If store expectancy reduces cognitive effort put forth during learning, one might expect the small effects observed here to compound into larger learning impairments across a longer period of study, like an academic semester. One might also expect these findings to persist even when material is restudied, as willingness to engage cognitively also predicts engagement in the type of challenging, engaged study strategies expected to promote effective learning (Cazan & Indreica, 2014; Evans et al., 2003; Weissgerber et al., 2018).

Moreover, students' motivation and goals related to the text could influence the source expectancy impairment observed in the current study. Our participants were likely unmotivated to read the texts presented in the context of the experiment. They were given no more reason to read the texts than being asked to do so in anticipation of a later test that did not affect any grade nor their compensation. Students would almost certainly have stronger incentives and motivation to engage with educational texts in a classroom setting. It is unclear how these baseline differences in motivation could interact with the store expectancy effects we observed. Demanding reading goals like intention to study, or even teach, can yield particularly strong memory and comprehension of the text (Coleman et al., 1997; Geiger & Millis, 2004; Guerrero & Wiley, 2021; Nestojko et al., 2014; Van den Broek et al., 2001). In a classroom setting, students with the demanding reading goal to learn in anticipation of a high-stakes exam could be less influenced by store expectancy than observed in the current study. On the other hand, the difference in motivation potentially brought about by store expectancy could be even more marked, moving students from high motivation to low, and resulting in even larger offloading-related impairments than observed here. Future work should examine these questions directly in classroom settings to elucidate how, when, and to what extent store expectancy could impair classroom learning.

The current study also used expository texts with a high reading level. Expository texts tend to be more difficult to read and less comprehensible than narrative texts (Clinton et al., 2020; Mar et al., 2021). Though we did not measure participants' reading level, we collected data from a sample of university students, who likely had higher reading ability than much of the population. Reading ability can predict the use of strategies which lead to successful learning from text (He, 2008), so skilled readers might be more affected by changes in study effort than unskilled readers. Specifically, shifts in strategy associated with store expectancy could be more consequential for skilled than unskilled readers, particularly if unskilled readers use ineffective reading strategies regardless of store expectancy. That said, skilled readers could instead be less affected by store expectancy if they process texts deeply regardless of store expectancy. Likewise, skilled readers may also need less time to effectively read a text than unskilled readers (Garguilo et al., 2026; Rayner, 1998). So, the differences in reading time that we observed might have even larger consequences for unskilled readers, who need more time to process a text. Future work should examine individual differences and text characteristics which might interact with source expectancy effects.

Overall, our findings align with studies observing store expectancy impairments, demonstrating the robustness of these effects even using materials one might expect to be protected from such impairment. This work invites additional inquiries investigating store expectancy effects with a wider variety of materials, including the types commonly saved externally using digital technology. As reliance on external storage systems increases, it is essential to identify the conditions that exacerbate or mitigate these offloading-related memory impairments. Expanding the scope of materials and contexts studied will develop our understanding of how store expectancy can influence subsequent memory, which could inform interventions aimed at mitigating these impairments in educational or everyday settings.

Notes

1. We counterbalanced the texts only across conditions which were manipulated as independent variables, holding other variables constant to reduce random variability. The function of the trust phase was to convince participants of the store expectancy manipulation. Participants also had access to one of the passages at test during the trust phase, so comparing test performance between the trust and experimental phases would not be particularly meaningful.
2. A mixed factorial ANOVA which included counterbalancing condition for order indicated no evidence of a main effect of condition order $F(1, 113) = 1.036$, $p = .311$, nor did this factor significantly interact with store condition $F(1, 113) = 0.028$, $p = .868$.
3. A mixed factorial ANOVA indicated no evidence of a main effect of condition order, $F(1, 97) = .146$, $p = .704$, nor did order with store condition, $F(1, 97) = 0.713$, $p = .401$.
4. The belief question was included to exclude participants who did not believe, understand, or attend to the store expectancy manipulation. The question, therefore, did not differentiate whether a participant reported that they did not believe they would have access to the passage due to an attentional or reading comprehension issue when reading instructions, scepticism toward the researchers, or participants' desire to report guessing our manipulation in hindsight. Only 33 participants did not believe the manipulation (22 of whom selected the ambiguous response "neither agree nor disagree"), so we did not think it appropriate to include this belief measure in our models as a covariate.

5. A mixed factorial ANOVA indicated a main effect of condition order, $F(1, 118) = 4.44$, $p = .037$ such that participants who Expected No Access first generally scored higher than those who Expected Access to the first passage. However, order did not interact with store condition, $F(1, 118) = 0.138$, $p = .711$.

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Author contributions

T.M.G.: Conceptualization, Data curation, Formal Analysis, Methodology, Visualization, Writing – original draft, O.D.R.: Conceptualization, Formal Analysis, Methodology, Writing – review and editing; Z.N.E.: Conceptualization, Methodology, Writing – review and editing J.S.S.: Conceptualization, Methodology, Project administration, Resources, Supervision, Writing – original draft.

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Data statement

All datasets and materials are available through the Open Science Framework: https://osf.io/enhqt/

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